Exploring DPO's Implicit Reward

Problem Statement

Direct Preference Optimization (DPO) derives its objective from the RLHF framework by showing that the optimal policy under KL-constrained reward maximization admits a closed form, which allows the reward model to be reparameterized in terms of the policy itself. The paper's central conceit, reflected in its title, is that a DPO-trained language model *is* a reward model: the quantity $\hat{r}(x, y) = \beta \cdot \log \left[\pi_\theta(y \mid x)/\pi_{\text{ref}}(y \mid x)\right]$ is claimed to play the same role as the explicitly trained reward model in the InstructGPT-style pipeline. This equivalence is exact only at the optimum and on the training distribution. What remains largely untested is whether the implicit reward **generalizes as a reward model**: does it rank unseen preference pairs as accurately as an explicit reward model trained on the same data, with the same base model and compute budget?

This matters practically. If the implicit reward generalizes well, DPO checkpoints could be reused as free reward models (for rejection sampling, data filtering, or evaluation). If it does not, the "secretly a reward model" framing holds only in a narrow, training-distribution sense — a meaningful qualification to a widely repeated claim.

Research question. Under a matched data and compute budget, how does the held-out preference accuracy of DPO's implicit reward compare to an explicitly trained Bradley–Terry reward model, in-distribution and under distribution shift?

Hypotheses. H1: the explicit reward model achieves higher pairwise accuracy on held-out, in-distribution preference pairs than the DPO implicit reward. **H2:** the gap widens under distribution shift (responses sampled from policies other than the reference, or drawn from a different prompt domain). A null result (implicit $\approx$ explicit) is equally informative: it would substantiate DPO's claim empirically at small scale.

Method

Models and infrastructure. Base model Qwen2.5-0.5B (Pythia-410M as fallback), fine-tuned with LoRA via HuggingFace TRL on a single GPU. All conditions share the same base model, SFT checkpoint, and preference data to keep the comparison budget-matched.

Data. Anthropic HH-RLHF, split into (a) training pairs, (b) held-out in-distribution test pairs, and (c) an out-of-distribution set constructed two ways: pairs from the other HH subset (train on *helpful*, evaluate on *harmless*), and pairs whose responses are generated by third-party models and labeled by a frozen gold judge (a larger instruction-tuned model).

Pipeline.

(1) **SFT:** fine-tune the base model on chosen responses to obtain the reference policy π_{ref}.
(2) **DPO:** train π_θ from π_{ref} on the training pairs; extract $\hat{r}(x, y)$ as the length-summed log-ratio scaled by β.
(3) **Explicit RM:** train the same base model with a Bradley–Terry pairwise loss on the identical pairs.
(4) **Evaluation:** score both models as preference classifiers on sets (b) and (c).

Metrics. Pairwise preference accuracy (primary); calibration of implied choice probabilities (ECE); Spearman correlation of each scorer against gold-judge ratings; length-controlled accuracy as a confound check (both scorers may partially proxy response length — we report accuracy on length-matched pairs to isolate this).

Ablations. *β sweep (DPO):* evaluate implicit-reward accuracy across $\beta \in \{0.05, 0.1, 0.3, 0.5\}$, since the implicit reward's scale and dynamics depend on β. *Preference budget:* train both scorers on $\{2\text{k}, 8\text{k}, 32\text{k}\}$ pairs to test whether the gap depends on data scale. *Training-progress curve:* evaluate implicit-reward accuracy at multiple DPO checkpoints, testing whether reward-model quality peaks before or after policy quality (connecting to overoptimization).

References

DPO (Rafailov et al., 2023); InstructGPT (Ouyang et al., 2022); Deep RL from Human Preferences (Christiano et al., 2017).